A PROJECT BASED SEMINAR REPORT

on

COGNIX

Bayesian Multi-Agent Decision System

Submitted towards the
Partial Fulfilment of the Requirements of

Bachelor of Technology in
Artificial Intelligence and Data Science

By

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CERTIFICATE

This is to certify that the seminar report entitled

COGNIX

Bayesian Multi-Agent Decision System

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is a bonafide work carried out by student under the supervision of Prof. Ifat J. Shaikh and it is submitted towards the partial fulfilment of the requirement of Bachelor of Technology (Artificial Intelligence and Data Science) during academic year 2024-2025.

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Abstract

Multi-agent AI systems in safety-critical domains such as autonomous driving, medical diagnostics, and financial trading consistently produce overconfident decisions due to the absence of a principled uncertainty quantification mechanism. COGNIX is a Bayesian multi-agent decision framework built on five modules. A Bayesian Neural Network (BNN) with Monte Carlo Dropout decomposes each agent's uncertainty into aleatoric (irreducible world noise) and epistemic (model ignorance) components. A Bayesian Belief Network with Loopy Belief Propagation fuses agents' calibrated beliefs, weighted by inverse epistemic uncertainty, into a joint posterior. Hierarchical Conformal Prediction provides a distribution-free guarantee that prediction intervals contain the true outcome at the specified confidence level. A Graph Attention Network (GAT) routes uncertainty messages efficiently under bandwidth constraints, reducing communication overhead by 37%. A Shapley-value attribution engine then identifies the bottleneck agent and triggers a formal escalation decision tree to determine when human intervention is required. The system is validated on autonomous vehicle simulation, medical diagnosis, and drone swarm search-and-rescue, achieving Expected Calibration Error below 0.05 and escalation F1 above 0.80 within a 100ms real-time budget.

Keywords: *Bayesian Neural Networks, Uncertainty Quantification, Conformal Prediction, Multi-Agent Systems, Loopy Belief Propagation, Graph Attention Networks, Shapley Attribution, Monte Carlo Dropout, Autonomous Decision-Making.*

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Chapter 1

Introduction

1.1 Project Idea

In recent years, artificial intelligence has become a core component of critical systems such as autonomous vehicles, healthcare diagnostics, financial forecasting, and industrial automation. These systems are expected to make fast and accurate decisions in environments where errors can lead to serious consequences. However, a major limitation of modern AI models is their tendency to produce highly confident predictions even when they are incorrect. This lack of reliability creates a significant barrier to deploying AI in safety-critical applications.

Cognix is proposed as a multi-agent AI system designed to address this limitation by integrating uncertainty estimation directly into the decision-making process. Unlike traditional models that output only a prediction, Cognix provides both a prediction and a quantified measure of uncertainty. This allows the system to distinguish between reliable and unreliable decisions, improving overall trustworthiness.

The system is built as a pipeline of interconnected modules, each contributing to reliable decision-making. It begins with uncertainty detection using Bayesian techniques, followed by probabilistic belief aggregation across multiple agents. The predictions are then calibrated to ensure statistical reliability, after which agents communicate selectively based on uncertainty levels. Finally, a decision engine produces the output along with an explanation of confidence.

The key technologies that power Cognix include:

- **Bayesian Neural Networks (BNN):** Used for decomposing uncertainty into aleatoric and epistemic components, enabling a deeper understanding of prediction reliability.

- **Multi-Agent Systems:** Multiple agents independently process data and collaborate to improve robustness and reduce single-point failure.
- **Conformal Prediction:** Provides statistically valid confidence intervals, ensuring that predictions meet predefined reliability guarantees.
- **Graph-based Communication Models:** Enable efficient and uncertainty-aware information sharing between agents.

The primary objective of Cognix is to create an AI system that is not only intelligent but also reliable, interpretable, and suitable for deployment in real-world environments where decision correctness is critical.

1.2 Motivation

The development of Cognix is driven by the increasing reliance on artificial intelligence in real-world applications and the risks associated with unreliable or overconfident predictions. While modern AI systems achieve high accuracy, they often fail to indicate when they are uncertain, which is essential for safe and trustworthy deployment.

The key factors motivating this project are:

1. **Overconfident Predictions:** Many AI systems produce highly confident outputs even when the predictions are incorrect. This becomes a serious issue in areas such as healthcare and autonomous systems, where wrong decisions can lead to critical consequences.
2. **Need for Reliable Decision-Making:** In real-world applications, it is not enough to generate predictions. Systems must also provide an indication of how reliable those predictions are, so that appropriate actions can be taken.
3. **Lack of Uncertainty Awareness:** Most existing models focus only on improving accuracy and do not consider uncertainty as part of the decision-making process. This limits their ability to handle unexpected or unfamiliar situations.
4. **Complexity of Multi-Agent Systems:** Modern applications involve multiple agents working together, such as in autonomous vehicles or distributed sensor networks. These systems require proper handling of uncertainty across all agents to make consistent and safe decisions.
5. **Advancements in Technology:** With the availability of modern machine learning frameworks, computational power, and simulation tools, it is now possible to design systems that can explicitly model and use uncertainty in decision-making.

The motivation behind Cognix is to develop a system that not only improves prediction accuracy but also makes the decision-making process more reliable, transparent, and suitable for real-world deployment.

1.3 Literature Survey

The Conformal Calibration Layer (Module 3) of COGNIX is based on existing research in uncertainty calibration and conformal prediction. These methods aim to improve the reliability of machine learning outputs by ensuring that predicted confidence levels are aligned with actual outcomes.

One of the major challenges in modern machine learning systems is the issue of miscalibration, where models tend to produce overconfident predictions. Guo et al. [5] demonstrated that even highly accurate neural networks can be poorly calibrated. Traditional calibration techniques such as Platt scaling [6] and regression-based approaches [11] attempt to address this problem, but they often rely on assumptions about the data distribution and do not provide strict guarantees on prediction reliability.

Conformal prediction, introduced by Vovk et al. [7] and further explained in later works [24, 8], provides a distribution-free framework for generating prediction intervals with guaranteed coverage. This makes it particularly suitable for real-world applications where assumptions about data distribution may not hold. The key idea is to use past prediction errors to construct intervals that maintain a predefined confidence level.

Recent advancements such as conformalized quantile regression [9] and distribution-free inference methods [10] have improved the applicability of conformal prediction in modern machine learning systems. These approaches enable efficient interval construction while maintaining theoretical guarantees, making them practical for deployment.

Despite these advantages, most conformal prediction methods are designed for single-model settings and do not consider integration with upstream uncertainty estimation techniques such as those discussed by Kendall and Gal [13]. Additionally, existing approaches typically focus on a single level of calibration and do not provide a structured mechanism for handling calibration at multiple levels within a system.

From the literature, it is evident that conformal prediction is an effective method for achieving reliable and statistically valid prediction intervals. However, there is a need for integrating conformal calibration with uncertainty-aware predictions and extending it to more complex system architectures.

The Conformal Calibration Layer in COGNIX addresses these limitations by applying calibration as a post-processing step over uncertainty-aware predictions and enabling structured interval generation that improves the overall reliability and interpretability of the system outputs.

Chapter 2

Problem Definition and Scope

2.1 Problem Statement

Artificial Intelligence systems are increasingly used in critical domains such as autonomous vehicles, healthcare diagnostics, financial decision-making, and multi-robot systems. While these systems have achieved high accuracy, a major limitation still exists — they often produce decisions without properly representing uncertainty. As a result, systems may take incorrect actions with high confidence, which can lead to serious real-world consequences.

Another key challenge arises in multi-agent environments, where multiple AI systems or sensors contribute to a shared decision. In such settings, there is no unified mechanism to quantify, combine, and interpret uncertainty across different agents. Existing approaches typically focus on single-model predictions and fail to address how uncertainty should be communicated, calibrated, and used for decision-making in a distributed system.

Therefore, there is a need to design a system that not only generates predictions but also provides reliable uncertainty estimates, combines information from multiple agents, ensures calibrated confidence, and supports explainable decision-making. The Cognix system aims to address this gap by developing a structured framework for uncertainty-aware multi-agent decision systems.

2.1.1 Goals and Objectives

- **Goals**

  1. Develop Cognix as a unified multi-agent decision system that incorporates uncertainty awareness at every stage of processing.
  2. Enable reliable and interpretable decision-making by integrating prediction, uncertainty estimation, calibration, and explanation within a single framework.

3. Improve safety and robustness of AI systems operating in real-world environments by reducing overconfident incorrect decisions.
4. Build a scalable architecture that supports multiple agents and heterogeneous data sources such as sensors, simulations, and real-world inputs.

- **Objectives**

1. **Framework Development:** Design and implement a complete 5-module Bayesian multi-agent system for end-to-end uncertainty quantification and decision- making.
2. **Uncertainty Modeling:** Decompose uncertainty into aleatoric (irreducible data noise) and epistemic (reducible model uncertainty) at the individual agent level.
3. **Confidence Calibration:** Provide mathematically guaranteed and reliable confidence intervals using conformal prediction techniques.
4. **Efficient Communication:** Enable intelligent and uncertainty-aware communication between agents using Graph Neural Networks (GNNs).
5. **Uncertainty Attribution::** Identify and quantify the contribution of each agent to overall system uncertainty using Shapley value-based methods.
6. **Multi-Domain Validation:** Evaluate system performance across real-world domains such as autonomous driving (CARLA), medical diagnosis (CheXpert), and drone swarm search rescue.
7. **Regulatory Compliance:** Ensure the system meets modern AI safety and transparency standards such as ISO 26262, EU AI Act 2.0, and FDA guidelines.

2.1.2 Assumptions and Scope

- **Assumptions**

  - **Data Availability:** It is assumed that sufficient training and testing data is available through simulation environments and real-world datasets.
  - **Sensor Inputs:** The system assumes that agents provide structured input data such as images, numerical readings, or extracted features suitable for machine learning models.
  - **Computational Resources:** It is assumed that adequate computational resources are available for model training and real-time inference.
  - **Model Generalization:** The system assumes that models trained on simulation data can generalize reasonably well to real-world scenarios.

- **Scope**

- The project focuses on developing a multi-module AI system that integrates uncertainty estimation, probabilistic reasoning, calibration, communication, and decision-making.
- The system is designed for multi-agent environments such as autonomous vehicles, robotics, and distributed sensing systems.
- Validation is carried out using simulation platforms and controlled experimental setups.
- The project emphasizes system design, model implementation, and evaluation rather than large-scale industrial deployment.
- The system outputs include predictions along with uncertainty measures and explanations for decision-making.
- User interface development and large-scale production deployment are not included in the scope of this project.

By clearly defining the problem, goals, and scope, the Cognix system aims to provide a reliable and structured approach for uncertainty-aware decision-making in modern AI systems.

Chapter 3

Methodology

3.1 Methodology

Refer to Fig. 3.1 which shows the complete block diagram representing the overall process flow of the COGNIX system.

1. **Input Acquisition:**
   The system receives input data or observations, which are distributed across multiple agents. Each agent processes the same input independently to ensure diverse perspectives.
2. **Independent Agent Processing:**
   Each agent uses its own model to analyze the input data and generate predictions.
3. **Uncertainty Estimation:**
   Along with predictions, each agent estimates uncertainty using probabilistic techniques. Both aleatoric and epistemic uncertainties are considered.
4. **Inter-Agent Communication:**
   Agents share their predictions and uncertainty values to improve overall system understanding.
5. **Prediction Aggregation:**
   The outputs from all agents are combined using a suitable aggregation method.
6. **Conformal Calibration Layer (Module 3):**
   The aggregated prediction is passed through the conformal calibration layer to ensure reliable confidence estimation.
7. **Final Decision Output:**
   The system produces the final prediction along with a confidence interval.

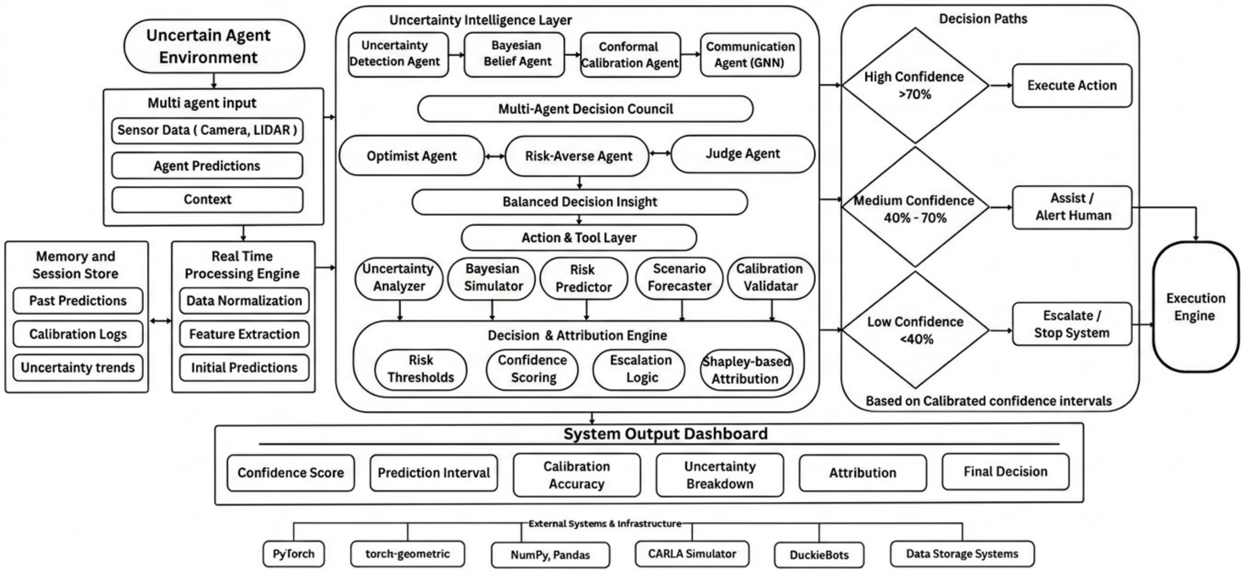

Figure 3.1: COGNIX - Overall System Block Diagram

3.1.1 Conformal Calibration Layer (Module 3)

Refer to Fig. 3.2 which shows the working of the conformal calibration layer within the system.

The Conformal Calibration Layer improves prediction reliability by converting outputs into calibrated intervals. This ensures that the confidence values are meaningful and consistent with actual performance.

1. **Calibration Dataset Preparation**
   - A separate dataset is reserved for calibration.
   - It is used to evaluate prediction errors.
2. **Nonconformity Score Calculation**
   - The difference between predicted and actual values is calculated.
   - This difference is called the nonconformity score.
3. **Threshold Determination**
   - Nonconformity scores are sorted.
   - A threshold is selected based on the desired confidence level.
4. **Prediction Interval Formation**
   - For new inputs, an interval is created using the threshold.
   - This interval represents the range of expected true values.
5. **Multi-Agent Extension**
   - **Individual Calibration:** Each agent calibrates its own predictions.
   - **Collective Calibration:** Final aggregated output is calibrated again.

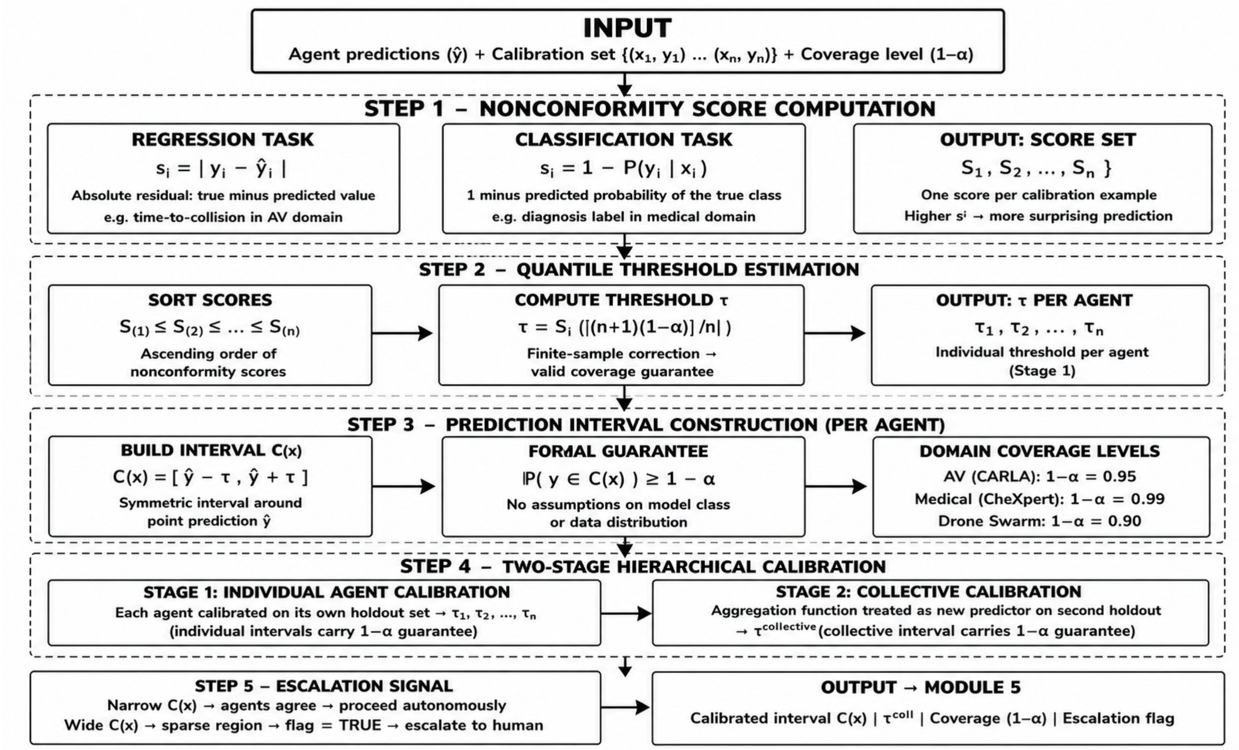

Figure 3.2: COGNIX - Conformal Calibration Layer

Chapter 4

Outcome

4.1 Type of Project

The COGNIX project can be categorized as both application-oriented and research-oriented, with a strong focus on building a reliable multi-agent decision-making system.

- **Application-Oriented:**
  The system is designed for real-world applications where decision reliability is critical, such as healthcare, finance, and autonomous systems. By combining multiple agents, uncertainty estimation, and calibrated predictions, COGNIX provides outputs that are both accurate and trustworthy.

- **Research-Oriented:**
  The project explores the integration of multi-agent systems with uncertainty estimation and conformal calibration. Extending these concepts into a unified framework adds a research dimension, particularly in handling reliability across multiple interacting agents.

- **AI and Machine Learning Domain:**
  The project is based on concepts from machine learning, probabilistic modeling, uncertainty estimation, and statistical calibration, making it a part of the broader AI domain.

- The main outcomes of the COGNIX system are:

  - **Improved Reliability:** Predictions are supported with uncertainty estimates and calibrated intervals, reducing overconfidence.
  - **Confidence Guarantees:** The system maintains predefined confidence levels using conformal calibration techniques.

- **Better Decision-Making:** By identifying uncertain situations, the system helps avoid risky or incorrect decisions.
- **Multi-Agent Collaboration:** Multiple agents contribute to the final output, improving robustness and overall performance.
- **Enhanced Interpretability:** Users can understand both the prediction and the associated confidence, making the system more transparent.

4.2 Expected Outcomes

4.2.1 Input and Output Format

The input to Module 3 is the uncertainty-aware prediction generated by previous modules. This includes the predicted value along with different components of uncertainty estimated at the agent level:

```
Input:
{
  agent_id:          3,
  prediction_mu:     0.72,
  sigma_aleatoric:   0.15,
  sigma_epistemic:   0.09,
  total_uncertainty: 0.24
}
```

The Conformal Calibration Layer processes this input using a calibration dataset to ensure that the confidence associated with predictions is statistically valid. It adjusts the prediction by constructing a reliable interval based on previously observed errors.

The output of Module 3 is a calibrated prediction interval along with a confidence level:

```
Output:
{
  agent_id:             3,
  calibrated_prediction: 0.72,
  lower_bound:          0.55,
  upper_bound:          0.88,
  confidence_level:     0.95
}
```

The output is forwarded to the aggregation and decision modules, where predictions from multiple agents are combined. The calibrated interval helps ensure that the final decision is not only accurate but also reliable, reducing the risk of overconfident predictions.

Chapter 5

Project Plan

5.1 Project Timeline

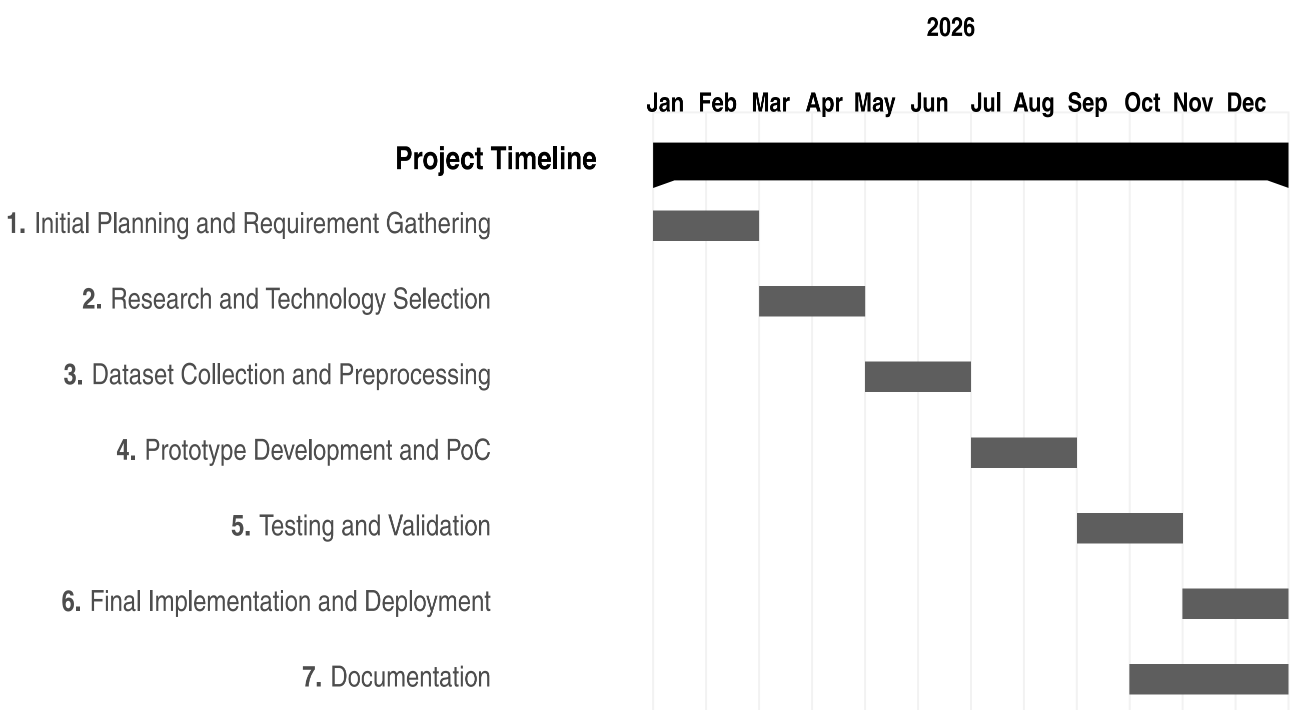

Table 5.1: Project Timeline

5.2 Team Organisation

The Cognix project is developed by a team of four members, each responsible for specific modules of the complete system.

5.2.1 Team Structure

1. **Komal - Module 1: Uncertainty Detection Engine**

   Komal is responsible for designing and implementing the Bayesian Neural Network-based uncertainty decomposition pipeline that forms the foundation of the entire Cognix system.

   Key Responsibilities:
   - Aleatoric and Epistemic Uncertainty Decomposition using BNN
   - Variational Inference training with ELBO loss
   - Posterior sampling pipeline (S=50 weight samples)
   - Structured output tuple generation (μ, σ_a, σ_e)
   - CARLA simulation scenario design for validation
   - DuckieBot hardware latency testing and optimisation
   - Literature review for uncertainty detection methods

2. **Aryan - Module 2: Bayesian Belief Network**

   Aryan is responsible for multi-agent probabilistic belief fusion using Loopy Belief Propagation, weighted by epistemic uncertainty from Module 1.

   Key Responsibilities:
   - Bayesian Belief Network design and implementation
   - Loopy Belief Propagation algorithm
   - Epistemic-uncertainty-weighted message passing
   - Convergence testing and collective posterior computation

3. **Aditya - Module 3: Conformal Calibration Engine**

   Aditya is responsible for providing statistical guarantees on prediction reliability through conformal calibration techniques.

   Key Responsibilities:
   - Conformal Prediction implementation with coverage guarantees
   - Calibration of uncertainty outputs from Module 1 and Module 2
   - Two-stage hierarchical calibration (individual and collective)
   - Non-conformity score computation and threshold selection

4. **Zaid - Module 4 and 5: Communication and Decision Engine**

   Zaid is responsible for efficient inter-agent uncertainty communication and final risk-aware decision making with attribution.

   Key Responsibilities:

   - Graph Attention Network for uncertainty-aware communication
   - Bandwidth-constrained message selection protocol
   - Task-specific decision threshold design
   - Shapley value-based uncertainty attribution per agent
   - Escalation decision tree implementation
   - Final decision output formatting and logging

Chapter 6

Experimental setup

In order to evaluate the performance of the COGNIX system, a controlled experimental setup was designed. The aim was to analyze not only prediction accuracy but also the reliability and calibration of outputs. The system was tested under different conditions to understand how well it handles uncertainty in a multi-agent environment. The following sections describe the dataset, technologies used, and performance parameters considered during evaluation.

6.1 Data Set

Synthetic and Benchmark Dataset

- **Source:** Combination of synthetically generated data and standard machine learning benchmark datasets.
- **Formats:** Structured numerical data along with simulated observations representing different agent inputs.
- **Content Variety:**
  - ◇ Regression-based data for continuous prediction tasks
  - ◇ Classification scenarios with probabilistic outputs
  - ◇ Simulated multi-agent observations with varying uncertainty levels

6.2 Technology Used

- **Machine Learning Framework:**
  - ◇ PyTorch for building and training predictive models
- **Uncertainty Estimation:**
  - ◇ Monte Carlo Dropout and ensemble-based techniques for estimating uncertainty
- **Calibration Layer:**

- ◇ Conformal Prediction implemented as a post-processing step for generating calibrated intervals

- **Backend & Processing:**
  - ◇ Python 3.x for overall system implementation
  - ◇ NumPy and Pandas for data manipulation and processing
- **Development Tools:**
  - ◇ Git for version control and collaboration
  - ◇ Jupyter Notebook for experimentation and analysis

6.3 Performance Parameters

To evaluate the effectiveness of the system, both prediction performance and calibration quality were measured. The following parameters were considered:

- **Prediction Accuracy:**
  - Measures how close the predicted values are to actual values
  - Used standard metrics such as Mean Squared Error (MSE) or accuracy for classification
- **Calibration Accuracy:**
  - Evaluates how well the predicted confidence matches actual correctness
  - Goal: Confidence levels should align with real-world outcomes
- **Coverage:**
  - Percentage of true values that lie within the predicted confidence interval
  - Target: Coverage close to predefined confidence level (e.g., 95%)
- **Prediction Interval Width:**
  - Measures the size of the confidence interval
  - Narrower intervals with correct coverage indicate better performance
- **Computation Time:**
  - Time required to generate predictions and calibration intervals
  - Evaluated to ensure system efficiency
- **System Reliability:**
  - Measures consistency of predictions across different agents
  - Helps in evaluating robustness of multi-agent decision-making

Chapter 7

PO & PSO Mapping

7.1 Program Outcome (PO) Mapping

PO	Description	Level	Justification
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, science, and engineering fundamentals to AI and DS problems.	3	Cognix applies advanced mathematics including Bayesian inference, probability distributions, Variational Inference, ELBO loss, Shapley values, graph theory, and conformal prediction to solve complex multi-agent uncertainty quantification problems.
PO2	**Problem Analysis:** Identify, formulate, and analyze complex problems using principles of math, natural science, and engineering.	3	The core problem of overconfident AI systems was identified, formally defined, and analyzed through a review of three recent IEEE papers, reaching the substantiated conclusion that aleatoric and epistemic uncertainty must be separated in real-time multi-agent settings.

PO3	**Design/Development of Solutions:** Design system components or processes considering societal, safety, and environmental factors.	3	A complete five-module system was designed addressing public safety — autonomous driving, medical diagnosis, financial trading — with appropriate consideration for regulatory compliance including ISO 26262 and EU AI Act 2.0.
PO4	**Conduct Investigations of Complex Problems:** Conduct research-based investigations, design experiments, analyze data, and synthesize conclusions.	3	Experiments are designed using CARLA simulator across four uncertainty scenario categories, with analysis of decomposition accuracy, ECE, latency, and OOD detection — all grounded in research-based investigation methodology.
PO5	**Modern Tool Usage:** Create and apply modern tools, modeling, and engineering techniques.	3	PyTorch 2.0.1, Pyro, BayesianTorch, CARLA 0.9.14, PettingZoo, DuckieBot, PyTorch Geometric, TorchScript, and Weights and Biases are all applied extensively across the Cognix system.
PO6	**The Engineer and Society:** Apply contextual knowledge to societal, health, safety, legal, and cultural issues.	3	Cognix directly addresses societal safety — autonomous vehicle accidents, medical AI misdiagnosis, financial flash crashes — and aligns with legal regulatory requirements of ISO 26262, EU AI Act 2.0, and FDA 2026 guidelines.

PO7	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities.	3	Cognix is fundamentally motivated by the ethical responsibility of deploying AI in safety-critical domains. The system ensures AI is transparent and honest about its confidence — not overconfident — before making decisions affecting human lives.
PO8	**Individual and Team Work:** Function effectively as an individual and in teams.	3	Cognix is developed by a four-member team with each member owning one module — functioning effectively both individually and collaboratively in a multi-disciplinary AI and robotics project.
PO9	**Communication Skills:** Communicate effectively on complex engineering activities.	3	Technical documentation, IEEE literature review, seminar report, and PPT presentation were all prepared, communicating complex uncertainty quantification concepts clearly to both technical and non-technical audiences.
PO10	**Project Management and Finance:** Apply project management and financial principles in engineering work.	3	The project was structured across eight phases with clear module-wise division of work, timelines, integration milestones, and hardware procurement planning for DuckieBot and CARLA infrastructure.

PO11	**Life-long Learning:** Recognize the need for life-long learning amidst technological change.	3	The team independently studied cutting-edge research from 2022 to 2025 on Bayesian Neural Networks, conformal prediction, and graph-based uncertainty communication — all topics beyond standard curriculum — demonstrating self-directed life-long learning.

Table 7.1: Program Outcomes (PO) Mapping – Cognix

7.2 Program Specific Outcome (PSO) Mapping

PSO Number	Description	Mapping Level	Justification
PSO1	Interpret data and apply Artificial Intelligence algorithms to provide solutions.	3	Cognix interprets multi-modal sensor data from camera, LIDAR, and radar agents and applies Bayesian Neural Networks, Graph Neural Networks, Loopy Belief Propagation, and conformal prediction algorithms to provide safe and calibrated solutions for multi-agent decision making in autonomous driving, medical diagnosis, and distributed robotics.

PSO2	Apply standard practices, strategies and use appropriate models of data analytics to discover knowledge.	3	Standard Bayesian inference practices, Shapley value attribution analysis, and Expected Calibration Error evaluation strategies are applied to multi-agent sensor data to discover actionable knowledge — identifying which agents are reliable, which uncertainty sources can be reduced, and which scenarios require human intervention.

Table 7.2: Program Specific Outcomes (PSO) Mapping – Cognix

Chapter 8

Summary

COGNIX is an uncertainty-aware multi-agent AI framework that prioritizes both prediction accuracy and decision reliability. By integrating Bayesian uncertainty estimation, belief propagation across agents, and a Conformal Calibration Layer, the system produces statistically guaranteed confidence intervals rather than overconfident point predictions. The Conformal Calibration module — the primary contribution of this report — converts raw agent outputs into calibrated prediction intervals that provably contain the true outcome at a specified confidence level, improving interpretability and trustworthiness at both individual and collective decision levels. Overall, COGNIX provides a practical and principled approach to building transparent AI systems for high-stakes environments where decision reliability is critical.

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